

Quantum Inspired Genetic Algorithm for Home Health Care Team Routing

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Quantum Inspired Genetic Algorithm for Home Health Care Team Routing

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Abstract. The main problem addressed by this paper is the routing of care worker teams. It consists in reducing the total cost and traveling time when visiting patients geographically distributed with satisfaction of care workers as well as patients. This problem is comparable to the real world multiple traveling salesman problem, patients should be considered as visited cities. This research work attempts to use a hybridization of quantum computing and genetic algorithm in home health care scheduling. The traditional evolutionary genetic algorithm has the disadvantages of poor local search ability and slow convergence. The advantage of the new quantum evolutionary approach is its ability to dig sensitively around reached minima to explore other possible optimization solutions without falling in as local minima. A comparative study between traditional genetic algorithm and quantum inspired genetic algorithm was carried out in the framework of this paper and showed a net superiority of quantum evolutionary approach in solving home health care routing problem.

Key words: home health care team routing; multiple salesman problem; genetic algorithm; quantum inspired genetic algorithm; evolutionary optimization.

1. Introduction

Nowadays, many health care centers and clinics adopt emerging Home Health Care (HHC) services. This includes nursing care, therapy and medical social services. In this kind of care systems, skilled practitioners provide care to patients in their homes under the direction of a physician. Care visit is usually planned before the day of visit; all participants (care workers and patient) in this visit are informed about its subject and its expected duration (starting and finishing times). The routes that care workers should take in their travel from the HHC center to the patient home and inversely should be well known and recognized as an optimal solution among many other solutions. Indeed, before starting the care worker their day trips, manager of the HHC center should solve this routing problem. The difficulty of scheduling the care worker route is related to necessity of synchronizing human and material resources (4). Minimization of the home health care global expenditures depends on the minimization of transportation costs that are known as the biggest sources of expenditure in home health care company activities (28). Moreover, other side constraints, such as patients' preferences and visiting time windows, have to be considered (11).

A good HHC organization should answer some objectives and constraints in order to satisfy both patient requirements and care worker preferences. In clinical or medical

center, HHC management rules presume that the same caregiver or nurse can provide the one or more home health care services in the same day for multiple patients. Patients are distributed in the same geographical area (such as in a town or a village). Care worker should perform patient assigned care tasks within this expected period of time (28). This time interval characterizes patient and defines the service time horizon usually named "time window" and during which a care activity can or must take place for a specific patient and at home.

We consider the HHC routing problem as an optimization problem using multi-objective functions with two kinds of constraints: patient satisfaction with less violation of time windows constraint and care worker satisfaction with less violation of the fairness principle of their workload. The main goal of HHC scheduling system is to determine and assign some care tasks (at patient homes) to home-care workers ("nurses" or "care givers") with respect to various side constraints, like an acceptable patient satisfaction and his preferred time-windows, employer workload fairness of care workers, and then a global minimization of transportation cost. Care workers start their day by checking in their daily visit plan provided by the manager of the HHC center. This plan points out which patients each care worker has to assist by visiting them at home and performing their required care services in a given time windows, specifying the expected start and end hours. Home health care workers start their daily trip from their workplace (clinical or care center) and travel for some distance to deliver care services to their assigned patients at their homes within a specified time window, after finishing all their assigned visits they return back to their workplace (18).

Traveling salesman problem (TSP) is a traditional routing problem. Its objective is to find the shortest route for a traveling salesman who, starting his trip from a specified city, he/she has to visit some other cities exactly once and then returns back to the original city (15). Several researchers consider the home health care problem as traveling salesman problem (TSP) where patients are compared to cities that salesman should visit without repassing by the same city. We consider the HHC problem as a multiple salesman routing problem (MTSP) where the main objective is to minimize the total traveled distance that leads to the minimization of traveling cost.

2. Related works

Home health care scheduling (HHCS) problem consists of designing a set of daily roadmaps for care workers which shows them routes to be used to visit assigned patients in order to provide them required care services or tasks. Patients must be treated at home and they are grouped by neighboring geographic area (10). The main HHCS

objective can be achieved by minimizing travel costs and respecting side constraints. Patient care trips must be planned beforehand in order to get satisfactions of both patients and care workers. Since we consider that HHCS problem should be solved similarly to vehicle routing problem, our attention has been addressed to different routing problem solving approaches. HHC scheduling literature reveals a global consensus about considering the home health care scheduling as a multiple salesman routing problem. Mohamed Cissé in (5) considers it as an extension of the vehicle routing problem (VRP) with unusual side-constraints that make the issues difficult to solve. Li Jun et al. in (15) proposed a genetic algorithm to solve multiple salesman problem. Yuan Zhi in (29) stated that this problem can be viewed as a combination of staff rostering problem and a vehicle routing problem with time windows (9,7,31). Kergosien Yannick et al. in (12) recognized that HHC scheduling problem is equivalent to a multiple traveling salesman problem with time windows with some specific constraints. He used an integer linear program with some technical improvements to solve it (1). Akjiratikarl C. et al. in (1) specifies that HHCS problem was solved using a Particle Swarm Optimization algorithm (population based evolutionary algorithm with local search algorithm). Begur S.V. in (2) solved nurse routing problem using an heuristic approach, De Angelis V. used linear program to solve this routing problem (7).

Frifita S. et al. in (8) think that HHC scheduling is an NP-hard problem and it is an extension of vehicle routing problem with additional complex constraints, particularly time windows and synchronization. They used a General Variable Neighborhood Search algorithm to solve it.

For solving the HHC routing problem Marcon Eric et al. in (16) specified that they used a dynamic multi-agent system using a set of reactive decision rules. They concluded that multi-agent paradigm is well suited to implementing the interactions among patients and caregivers. In their HHC multi-agent modeling, they used two types of agents: Patient agent and caregiver agent. A set of performance indicators was considered for the both caregivers and the patients. The main objective of their research work is to minimize the travel times for caregivers and to reduce the waiting times for the patient's home (16).

For addressing the home health care scheduling problem, Mutingi M. and Mbohwa C. in (20) proposed a group genetic algorithm (GGA). The approach utilizes the strengths of unique group genetic operators to address the group structure of the problem. They stated that good solutions were provided within reasonable computation times.

Few research works used quantum genetic algorithm as optimization technique to solve vehicle routing problem. Beheshti A.K. and Hejazi S.R. in (3) proposed the use of quantum genetic algorithm as integrated model for vehicle routing problem with delivery time cost. In the proposed algorithm, various operators such as observation process and gate operator were applied.

Several research works have been investigated to solve this multi-objective optimization problem using different techniques going from traditional optimization problem solving tools to evolutionary techniques. The present research paper aims to take advantage from the quantum genetic algorithm evolutionary optimization approach to solve care worker routing problem as a multiple salesman routing problem.

3. Home health care routing problem description

HHC routing problem can be approached to MTSP problem by considering each caregiver as salesman and patients as towns. He/She should route from one patient to another without passing by the same patient in the same schedule. The main goal of the management of HHC is to minimize the total cost incurred by caregivers in travelling from the point of origin to the patients and back to the point of origin. In home health care scheduling problem, patients are dispersed geographically and require possibly several care tasks. Multiple care staffs should be routed through some roads. Roads connect different patient locations. Each patient requires possibly several care tasks. A care task is characterized by a duration, an earliest and a latest starting times and one needed category of care workers (a nurse or a doctor or etc.), A care task can also require several care workers (4). The problem description given below is borrowed partially from three referenced papers (5,20,21) for HHC scheduling problem:

- The number of care workers is fixed to 3 for simplicity of application.
- Care workers are specialized in one or more care tasks. The number of care tasks is equal to 5.
- As restriction, each patient can be visited only once per day. This means that he/she can welcome only one care worker or team in the same day. The number of patients is equal to 9 for every working day.
- Care workers are assigned jobs according to patient requirements described in care tasks.
- All care worker jobs have the same priority.
- Visits of patient should start at 9:00 am and finish at 15:00 pm. A break time between 12:00 and 13:00pm is free from every caregiver engagement or visit.

- Every delay of care worker to reach patient at his preferable start time should be penalized. Similarly, arriving early or before the start time of a patient time window should be also penalized. The travel speed is not considered as parameter here.
- The location of patient homes is defined by Geo-localization (Latitude and Longitude coordination). For simplicity of our application, we use two coordinates X and Y.
- Transportation is the responsibility of HHC center, which is in charge of transportation of care workers from HHCC to patient or from one patient to another or from a patient to the HHCC.

When considering HHC routing problem as a multiple traveling salesman problem (MTSP), MTSP is defined as some salesman who should travel through a network of delivery points or map nodes. They start their delivery trips at the same store point and get back to it after visiting their destinations and accomplishing their planned delivery tasks. Multiple salesmen routing problem solving approach tries to find some routes which can be used by salesmen and ensure that each salesman can visit a delivery place only once per trip and the total cost of all salesman trips is well minimized (8,3). The number of salesman is referred to the number of routes collected. In HHCS problem, patients should replace cities in traditional MSTP and caregivers should be considered as salesman. As multiple salesman routing optimization problem, the total transportation cost in HHC scheduling problem should be minimized. The minimization of total cost can be translated simply into the minimization of total traveled distance as:

$$\min \sum_{i=1,nc} \sum_{j=1,np} \gamma (d_{j,j+k} + \lambda + v) \quad (1)$$

Where :

- $d_{j,j+k}$ is the distance between HHCC and patient, or distance from patient to patient or distance from patient to HHCC.
- v is the penalty for arriving early and λ is the penalty for arriving late.
- v is equal 0 if caregiver arrives in time. Elsewhere, it is positive value is used to penalize the minimization process.
- λ is equal 0 if caregiver arrives in time. Elsewhere, it is positive value. Elsewhere, it is positive value is used to penalize the minimization process.
- γ is equal to 1 if patient is visited by caregiver, 0 elsewhere.

4. Genetic algorithm

Based on Darwin biological evolution principle of natural selection, John Henry Holland, the American scientist and professor of psychology and electrical engineering

and computer sciences, developed a genetic algorithm optimization method and he presented it in his book "Adaptation in Natural and Artificial Systems" as an abstraction of natural selection. The GA represents a set of optimization problem solutions as population of "chromosomes" (e.g., strings of ones and zeros, or "bits"). Each individual element of population is named "chromosome" which consists of vector or set of characteristics or "genes" (e.g., bits: 0 or 1). The transition from one to a new optimal population was abstracted using a fitness function and some genetics-inspired operators of crossover, mutation.

The selection operator chooses the chromosomes in the population which are able to reproduce. Selection principle is directed by a maximization process, for a problem with a minimum solution as its optimization objective, it needs to turn it to a maximum one minimization. The fitness function serves as a selection filter; it is used in a GA to judge the chance that an individual chromosome can be selected into the next generation.

A fitness function should be well defined at the beginning of evolutionary optimization process. Sometimes, it can be considered as a limiting factor to the efficiency of evolution, a small adjustment of the fitness function can have a significant effect on the convergence of GA. Several research works were investigated in developing GA selection strategies based on the principle of Darwin natural selection process "the larger fitness the better individual" (17).

Crossover exchanges subparts of two chromosomes, similarly to biological reproduction, by recombination between two single-chromosome organisms. The first subpart of the first chromosome will go with the second subpart of the second chromosome and the second subpart of first chromosome will complete the first subpart of the second chromosome. Crossover point defines the allele (bit) location on the chromosomes, which divides the chromosome into two sub-parts. In GA, the crossover point is defined using a crossover probability. The crossover probability gives the percent of population or individuals that will participate in the crossover operation.

Mutation consists of changing the value of a given chromosome position (allele) by its complement as a binary number (1 is changed to 0 and 0 is changed to 1). It randomly changes the allele values of some locations in the chromosome. GA mutation allow avoiding local minima by preserving diversity of population. During an evolutionary process, if the diversity of population is not guaranteed, chromosomes may become similar to each other, and thus evolution becomes slow or even stop. (17).

In Conventional GA, chromosomes of initial population are randomly constructed. The evolution of the population from one generation to another is directed by a fitness evaluation function, an individual or chromosome is selected to reemerge from the current generation to the next based on its fitness function value. New populations are

generated by natural selection, crossover and mutation. This evolutionary process continues until the termination conditions are satisfied (9).

Even though, GA has better adaptability and wide accommodation, it also has the disadvantages of poor local search ability and slow convergence (26). To resolve these weaknesses, the present paper proposes the use of quantum genetic algorithm.

5. Quantum Genetic Algorithm

Quantum mechanics is a branch of physical science that deals with the very small particles (atoms and subatomic particles). Usually the behaviors of these particles are unpredictable and uncertain and it is difficult to know early their trajectory. Uncertainty is described by the probability that a part of atom will evolve in one state or another. Quantum computation is inspired from quantum mechanics. Quantum computing is based on the ability of one bit to have two different values in the same time: zero (0) and (1) instead of having, the traditional binary digit values zero (0) or (1). This new ability provides IT technology some new advantages compared to the traditional computing methods. The parallelism that the quantum computing brings reduces obviously the algorithmic complexity. Such parallelism ability allows new IT technology to reach high processing speed and big memory storing capacity. Quantum inspired computing is characterized by the use of some principles of quantum mechanics such as principle of superposition of states and principle of indeterminacy (or non-predictable future state) (19) (22).

Quantum Genetic Algorithm (QGA) is the result of combination of traditional genetic algorithm evolutionary concepts and quantum computing mechanism. Quantum computing is a new field in computer science which takes its origins from the foundations of the quantum physics. It was got enormous attention because of its small population size, higher ability to find the global optimal value, fast convergence rate and time-saving (26). It is based on Qubits and states superposition of quantum mechanics. Quantum genetic algorithm has excellent convergence speed and global search ability, and the performance of the algorithm will not be affected with a small population size (21) (30).

5.1 Quantum Bits and Quantum Algorithms

The smallest unit of information in computers is named "bit" and it is able to be in only two different states: zero (0) or one (1). However, in quantum computing, the quantum analogue of the bit is named "Qubit". The Qubit is can be in two states denoted by $|0\rangle$ or $|1\rangle$ or a linear superposition of these two states. It is defined by:

- a bit of information that can be zero and one simultaneously.

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assigned to every quantum superposition state of each chromosome of the initial population (27)(12)(23).

5.4 Quantum Rotating Gate

The smallest information unit in the computer is called bit with only two possible values one (1) or zero (0). However, the basic unit for information storage of quantum computer is qubit (27). A qubit is a sequence of n bits maintained as n -bits quantum register in a mixed state $|\psi\rangle$, which is the coherent superposition state of 2^n ground states. Contrarily to regular computer, where two bits can represent four states (00, 01, 10, or 11), the quantum computer qubits, because of superposition, can represent all four states at the same time. The relationship between superposition state and ground states is given by equation 3:

$$|\psi\rangle = \sum_i c_i |\psi_i\rangle, \quad (3)$$

Where c_i represents the probability amplitude of the state $|\psi_i\rangle$; $|c_i|^2$ represents the probability to collapse to the ground state $|\psi_i\rangle$ when the state $|\psi\rangle$ is measured; c_i should satisfy the condition (12):

$$\sum_i |c_i|^2 = 1, \quad (4)$$

For example, a quantum system of a single qubit is given by equation 5:

$$|\psi\rangle = c_1|0\rangle + c_2|1\rangle \text{ and } |c_1|^2 + |c_2|^2 = 1. \quad (5)$$

Therefore, quantum genetic algorithm uses the probability amplitude $|c_i|^2$ of qubits to encode chromosome and it uses quantum rotating gates for chromosomal updated operation. The selected quantum chromosomes to persist from one generation to another are determined by the optimal individuals of the parent (5). Indeed, when applying a quantum rotation gate to the quantum superposition states, the probability amplitudes of all ground states can be changed, so that the corresponding chromosomes will be updated (9). Quantum gate is a variation operator, by which the qubit is updated to satisfy the normalization condition: $|c_1|^2 + |c_2|^2 = 1$.

The application of some quantum gate to the current superposition state $|\psi\rangle$ leads to the update of values of c_1 and c_2 , and the condition $|c_1|^2 + |c_2|^2 = 1$ with new c_1 and c_2 values still preserved. Thus, the probability amplitude of the ground state $|1\rangle$ decreases if the probability amplitude of the ground state $|0\rangle$ increases ($|c_1|^2$ and $|c_2|^2$ are two complementary values to 1).

The quantum rotation gate $U(\theta)$ is represented by a matrix in equation 6 and rotation angle variation is given by figure Fig. 3. Quantum rotation gate will be used to update the chromosome as expressed by equation 7.

$$U(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \quad (6)$$

Where θ is the rotation angle that is illustrated by figure Fig 3. The table Table1 shows some example of $U(\theta)$ matrix.

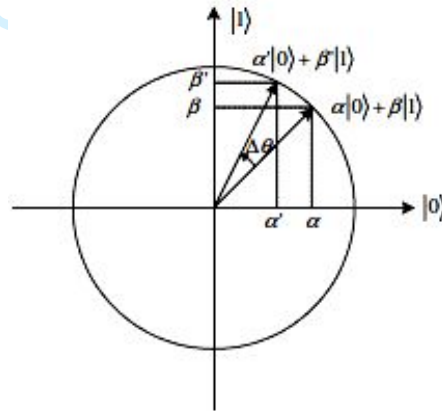


Fig. 3. Quantum Rotation Gate Qubit Update Operation (24)

Table 1. Examples of Quantum Rotation Gate Matrix (23).

Quantum qubit update is given by the equation 7.

$$[\alpha'_i \ \beta'_i]^T = U(\theta_i) [\alpha_i \ \beta_i]^T \quad (7)$$

Where:

- The two vectors $[\alpha_i \ \beta_i]^T$ and $[\alpha'_i \ \beta'_i]^T$ are the probability amplitudes of the i -th qubit in chromosome before and after the quantum rotating gates updating, respectively.
- $\theta_i = (\alpha_i, \beta_i) * \Delta\theta_i$, (α_i, β_i) and $\Delta\theta_i$ represents respectively the rotation angle direction (+ or – or $\pm$) and value of rotating angle. The lookup table is presented in Table 2.
- $\Delta\theta_i$ represents the rotation angle of each qubit toward either 0 or 1 states depending on its sign (α_i, β_i) . It determines also the speed of convergence, and its sign determines the direction of convergence (5).
- $\Delta\theta_i$ is considered as the step size rotation angle and it should be designed in compliance with the application problem (11).

Table 2 summarizes some recommended adjustment strategies of rotation angle variation $\theta_i = (\alpha_i, \beta_i) * \Delta\theta_i$, where:

- x_i is the i^{th} bit of the current chromosome;
- b_i (best i) is the i^{th} bit of the current optimal chromosome;
- $f(x)$ is the fitness function;

The comparison of the fitness of the currently measured value $f(x)$ of the individual q_{tj} with the fitness of the current optimal individual $f(i)$, the probability amplitude (α_i, β_i) evolves toward two directions (5):

- if $f(x) > f(b)$, then adjust the corresponding qubits of q_{tj} , making the probability amplitude (α_i, β_i) evolves toward the direction that is favorable to the emergence of x_i .
- Conversely, if $f(x) < f(b)$, then adjust the corresponding qubits of q_{tj} , making the probability amplitude (α_i, β_i) evolves toward the direction that is favorable to the emergence of best fit b_i

Table 2. Rotation Gate Angle Selection Strategy (2,11)

5.5 Evaluation of Solutions

This stage consists of quantifying the quality of each quantum chromosome in the population using a fitness function. Individuals with the highest fitness values will participate in the next step (or generation) of evolutionary (or reproduction) process.

6. Quantum Inspired Genetic Algorithm

Another alternative was developed and named "Inspired of QGA Quantum" by applying know traditional genetic algorithm operations (crossover, mutation) to quantum genetic algorithm. The quantum-inspired evolutionary algorithms are located in the intersection of two subareas of computer science, quantum computing and evolutionary computing. This approach does not require a functional quantum computer (20).

6.1 Chromosome Measurement

When facing real case applications of superposed states of qubits, the observation of each qubit leads us to convert it into binary chromosome as illustrated in Figure 5. To enable the evaluation of each quantum chromosome, we use a simple measure or conversion way by assigning 0 or 1 for each qubit using the following algorithm named collapse (or measurement):

```

Algorithm collapse
Begin
  Loop i=1 to n    /* n is the size of chromosome */
    Loop j=1 to m /* j is of gene: for each  $q_j^t (\alpha_{ij}^t, \beta_{ij}^t)$  */
       $\alpha \leftarrow \alpha_{ij}^t$ ;
      rand(r); /* generate a random value  $\in [0,1]$  */
      if  $r \geq |\alpha|^2$ 
         $\alpha_{ij}^t \leftarrow 0$ ; /* assign 0 to  $\alpha_{ij}^t$  of qubit  $q_j^t$  */
         $\beta_{ij}^t \leftarrow 1$ ; /* assign  $(1 - (\alpha_{ij}^t)^2)^{1/2} = 1$  to  $\beta_{ij}^t$  of qubit  $q_j^t$  */
      else
         $\alpha_{ij}^t \leftarrow 1$ ; /* assign 1 to  $\alpha_{ij}^t$  of qubit  $q_j^t$  */
         $\beta_{ij}^t \leftarrow 0$ ; /* assign  $(1 - (\alpha_{ij}^t)^2)^{1/2} = 0$  to  $\beta_{ij}^t$  of qubit  $q_j^t$  */
      end;
    end Loop /* end of gene loop */
  end loop; /* end of chromosome loop */
End collapse /* end of algorithm measurement */

```

When applied to a chromosome measurement of three genes, Algorithm Collapse gives the result described by table 3.

Table 3. Example of Chromosome measurement of 3 genes

6.2 Quantum Crossover

Quantum genetic algorithm uses two quantum operators: crossover and quantum mutation. Quantum Crossover operator is similar to natural reproduction that allows structural information exchange between individuals. Quantum crossover probability allows randomly selection of portion of individuals from the population to be involved in crossover operation to participate in the exchange of chromosome structure information.

6.3 Quantum Mutation

The role of quantum mutation is to sensitively disturb the evolutionary status of some individuals (chromosomes or solutions) in the current population and improve the local search capabilities. The value of qubit is different from the ordinary bit value (0 or 1) by the superposition of two different states ($|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and $|\alpha|^2 + |\beta|^2 = 1$) as described by equation 2. Thus, the quantum mutation consists of replacing $|\alpha|^2$ and $|\beta|^2$ values by their complements to 1. Mutation probability P_e allow to select a quantum bit or a part qubit gene, and realize the mutation through quantum NOT gate (25).

6.4 Quantum Catastrophe

As every optimization method, QGA sometimes falls into local optimum. Catastrophe happens when the optimization process becomes not able to over fit local extremum and the best individual of the previous generation does not change in successive generations. Evolutionary process should be disrupted in order to allow the optimization process to leave the stationary stage. The essence of this quantum evolutionary disruption is to keep the best individual in the population as the first chromosome of next generation. (26)

7. Quantum inspired Genetic Algorithm for HHC Routing.

The schedule preparation starts by collecting one day or more before patient home health care requests. Patient calls Home Health Care Center to make an appointment. If he/she is registered then administrator can add his/her request to the nearest available appointment. After gathering some home health care requests, care workers should be assigned to patient based on their qualification and competence to help of patients. Patient record should contain sufficient information about the patient health state. Before starting the schedule preparation, a qualification matrix should be drawn to show which care task can be performed by given care worker. Table 4 illustrates a qualification matrix and table 5 shows patient requirements in term of care tasks. The

following home health care example assumes that there are three care works qualified in some care tasks among a set of five care tasks. The number of patients is equal to nine (9).

Table 4. CQM Care Qualification Matrix of care workers

Table 5. MPR Matrix of patient requirements

Where the five care tasks are T1, T2, T3, T4, T5 and the nine patients are P1, P2, P3, P4, P5, P6, P7, P8 and P9.

Home health care schedule is established by defining some possible and optimal paths through an assumed geographical network of patients. In the same day, a care worker cannot pass twice by the same patient. Route scheduling deals with creating routes for care workers which daily travel to answer patient requests. Each patient is visited by once per day. The home care plan is daily prepared to allow some care workers or teams to visit patients at their different locations to accomplish different and required care tasks. Problem solving process is comparable to multiple salesman problem solving. Table 6 shows a symmetric matrix of patient distances. The related network of patients of figure Fig 4 shows three examples of routes or care worker paths.

Table 6. Symmetric Matrix of distances between patients

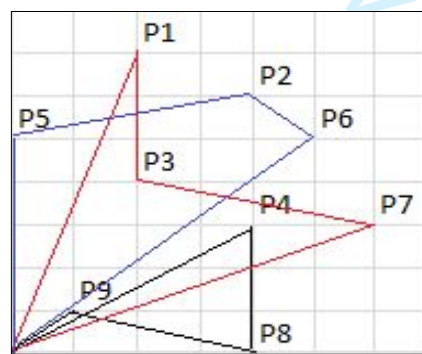


Fig. 4. sample of possible routing of 4 care teams

7.1 Chromosome Encoding

Given nc the number of care workers ($nc=3$) that should visit np distant patients ($np=9$). We encode the basic data of this routing problem as a chromosome encoding vector where each element (cell) is defined by the care worker identifier (id) or number which is in charge by visiting this patient. A solution is described by a chromosome with 9 digits, every digit is assigned to one patient and the same care worker ID could appear

7.2 Conventional GA Chromosome Encoding

[illegible]

7.3 Quantum GA Chromosome Encoding

α_{00}	α_{01}	α_{10}	α_{11}	$\cdot$	$\cdot$	$\cdot$													α_{110}	α_{11}
β_{00}	β_{01}	β_{10}	β_{11}	$\cdot$	$\cdot$	$\cdot$													α_{110}	α_{11}

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7.4 fitness

Quantum fitness function starts by converting chromosome binary values into decimal values that represent the ID of care workers. The index of chromosome element represents the ID of the patient. HHCS chromosome shows for every patient, the care workers who are intended to visit him/her. The fitness is computed based on the summation of distances traveled by every care workers. A care worker should start his/her day from the HHC center and pass by all patients assigned to him/her. At the end of his/her trip, he/she should get back to the center. Fitness function is given by equation 8.

$$\text{Fitness} = \sum_{1 \leq i \leq n} (D(C, P_{i1}) + \sum_{1 \leq j \leq m-1} D(P_{ij}, P_{ij+1}) + D(P_{im}, C)) \quad (8)$$

Where:

- n is the number of care workers or teams
- m is the number of patients visited by the care worker.
- (P_j, P_{j+1}) is the distance between two consecutive patients of the a care work team i
- $D(C, P_{i1})$ is the distance from the care center C to the first visited patient P_{ij} .
- $D(P_{im}, C)$ is the distance from the final visited patient P_{ij} by the care worker team I to the care center C .

7.4 Algorithm

The evolutionary manipulation of quantum genetic algorithm uses quantum-rotating gates to change the probability amplitude. The performance of quantum genetic algorithm can be improved by modifying the amplitudes of individuals or chromosomes in the sense of the value of the best solution (5). Therefore, the construction of Quantum rotating gates is the key issue of quantum genetic algorithm (1). The rotation of individual's amplitudes is performed by quantum rotation gates. The value of the rotation angle $\Delta\theta$ has to be selected so that to avoid premature convergence. Quantum gates can also be designed in accordance with the specificity of the on hand problem. As stated by Zakaria Ladoudi et al. in (14), when compared with the conventional genetic algorithm, quantum genetic algorithm applies the probability amplitude of qubits to encode chromosome and uses quantum-rotating gates to perform chromosomal updated operation. It consists of moving the state of each qubit $q_j^t (\alpha_{ij}^t, \beta_{ij}^t)$ as described by the figure Fig. 1.

Similarly, to conventional GA approach, a quantum inspired GA optimization process starts by having an initial population with randomly generated chromosomes. This

initial population has to evolve to an optimal solution by applying both conventional GA operators (crossover, mutation and selection) and quantum computing inference process using $\cos(\theta_i)$ of rotation angle gate θ_i instead of α_i . The Quantum Inspired Genetic Algorithm is defined as a mixing of both GA and QGA. GA algorithm known operators are added in basic version of QGA :

- Quantum crossover: it is similar to traditional GA crossover function but should be applied to qubit chromosome instead of applying to binary chromosome.
- Quantum Mutation: it is similar to traditional GA mutation function but should be applied to qubit chromosome instead of applying to binary chromosome.

Two other functions should be considered in QGA version, which are:

- Collapse or measurement function is used to convert chromosome qubit into binary chromosome.
- Disaster function is used to allow QGA to leave the local minima when it is falling in.

Quantum Inspired Genetic Algorithm (QIGA) that we propose to be used as quantum evolutionary inspired optimization algorithm as described by the algorithm HHC_QIGA and the chart of figure Fig.7.

Algorithm HHCS_QIGA

TYPE Bin_Chrom: array of binary with size=number of patients /* it describes the assignment care workers X patients*/

TYPE Q_Chrom: array of quantum (α, β) with size=number of patients /* it describes the assignment care workers X patients*/

Input:

P_Dist: array that stores patient distances;

Initialize:

sizepop: Size of population;

Q_P: array of sizepop structure of {fitness, Bin_Chrom, Q_Chrom};

B_fit_Chrom : Best fitness Quantum chromosome of Q_P;

begin

while (not termination criterion) do

$t \leftarrow t+1$

loop $i=1$ **to** sizepop

 Qcollapse(Q_P(t),i); \COMMENT{measurement function which translate quantum chromosome into binary chromosome}

 Q_P.fitness(t,i) $\leftarrow$ fitness (Q_P(t),i); /* fitness calculs*/

loop $j=1$ **to** i \COMMENT{research of best fitness: the minimum total of traveled distances}

If (B_fit_Chrom > Q_P.fitness(t,i)) **then**

 B_fit_Chrom = Q_P.fitness(t,i);

end if

end loop /* end for j*/

end loop /* end for i*/

 Qcross(Q_P.fitness(t));

 Qmutation(Q_P.fitness(t));

 Qgate (Q_P.fitness(t), B_fit_Chrom); \COMMENT{update of quantum population}

end while

end algorithm HHCS_QIGA

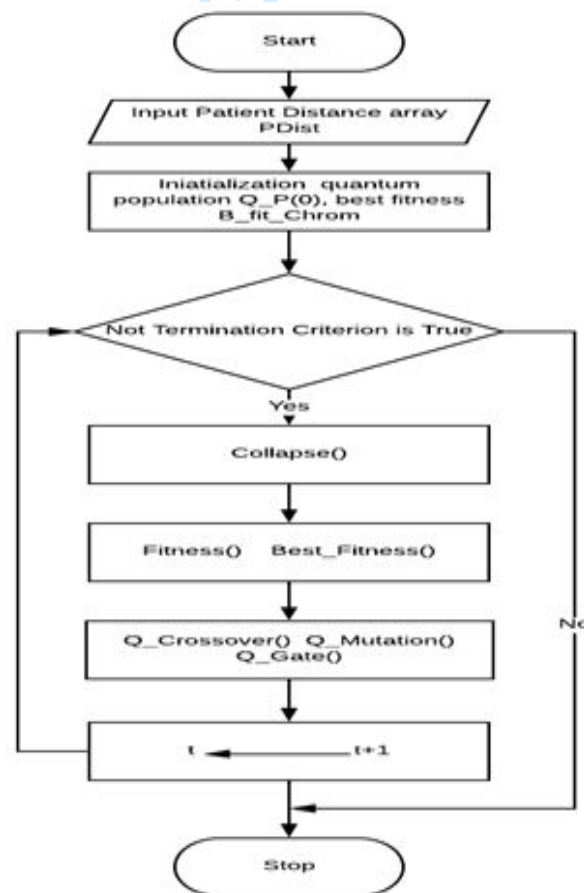


Fig. 7. Chart of Quantum Inspired Genetic Algorithm

8. Results and Discussion

The two algorithms GA, QIGA were implemented using MATLAB programming language and a computer of intel processor Intel R Core™ i5-23210M CPU of 2.5 GHZ of 4 G RAM under windows 64 bits operating system. We used the same data set presented before of 9 patient and 4 care worker teams and a distance matrix during the fifty iterations, The Quantum Inspired Genetic Algorithm (QIGA) was shown a net superiority compared to Genetic Algorithm (GA). It was more stable and speedy to reach the best optimal solution. Figure Fig 8 illustrates this superiority of QIGA against GA. In this QIGA version, we took advantage from both traditional evolutionary operators (crossover and mutation) and the speed of the new Quantum Genetic Algorithm with angle variation $\Delta\theta$ of rotation gate equal to 0.025π .

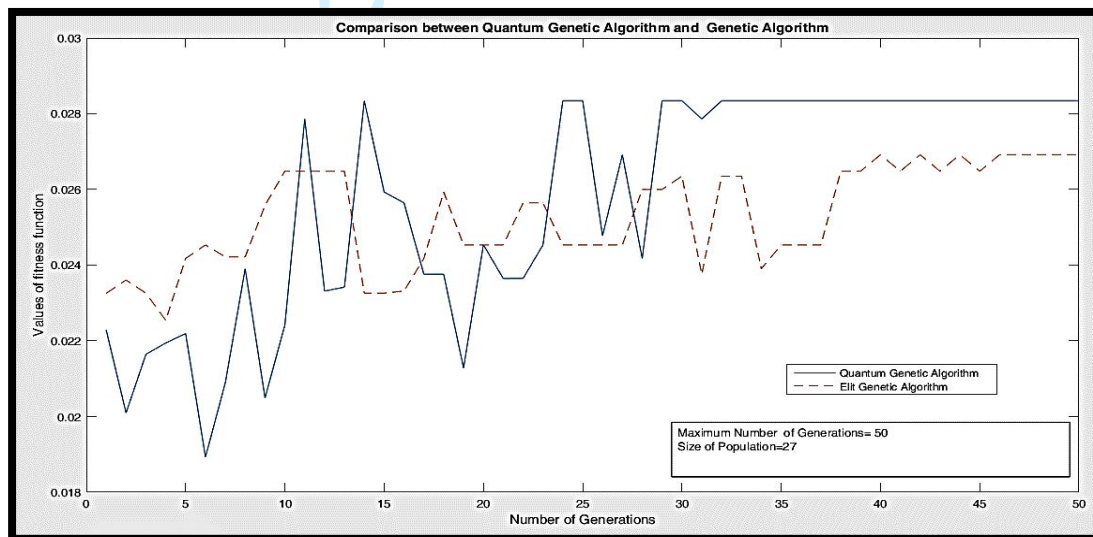


Fig. 8: Comparison between GA and QIGA

9. Conclusion

The present paper solved the HHC routing problem using quantum inspired GA. The basic version of QGA was not performant when searching the minimum of total travel distances of care worker teams visiting geographically dispersed patients. A new quantum inspired genetic algorithm (QIGA) was developed. It took advantage from the evolutionary operators (crossover and mutation) of traditional GA and the high speed processing dedicated to Quantum GA. The new quantum inspired genetic algorithm QIGA showed net superiority when reaching minimum, compared to traditional GA. To improve the performance of QIGA and take consideration of time windows constraints and the multiple visits of the same patient by different care workers, a new research investigation will be conducted in the nearest future

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Table 1. Examples of Quantum Rotation Gate Matrix (23).

Name of Gate	Rotation Gate Matrix	Role of Gate	Description of Gate result
Hadamard Gate H	$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	Single qubit gate. It maps the basis states as follows : ▪ $0\rangle$ to $\frac{ 0\rangle + 1\rangle}{\sqrt{2}}$ ▪ $1\rangle$ to $\frac{ 0\rangle - 1\rangle}{\sqrt{2}}$	It is a combination of two rotations, π about the X-axis followed by $\frac{\pi}{2}$ about the Y-axis. There are equal probabilities to become 1 or 0 and create a superposition ($\alpha^2 = \beta^2$).
The Pauli-X gate NOT Gate (23)	$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	Single qubit gate. It exchanges the probability amplitude of $ 0\rangle$ and $ 1\rangle$: ▪ $0\rangle$ is replaced by $1\rangle$ ▪ $1\rangle$ is replaced by $0\rangle$	There is a rotation around the X-axis by π radians. It maps of $ 0\rangle$ to $ 1\rangle$ and $ 1\rangle$ to $ 0\rangle$. It is sometimes called bit-flip
Phase Shift Gate	$R_\phi = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{bmatrix}$ Where ϕ is the phase shift.	Single-qubit gate. The basis state $ 0\rangle$ stills unchanged and the state $ 1\rangle$ is mapped to $e^{i\phi} 1\rangle$.	Some common examples are: ▪ $\frac{\pi}{2}$ Phase Shift Gate ($\phi = \frac{\pi}{2}$) ▪ $\frac{\pi}{4}$ Phase Shift Gate ($\phi = \frac{\pi}{4}$) ▪ $\frac{\pi}{8}$ Phase Shift Gate ($\phi = \frac{\pi}{8}$)

Table 2. Rotation Gate Angle Selection Strategy (2,11)

x_i	b_i	$F(x) < f(b)$	$\Delta\theta_i$	α_i, β_i			
				$\alpha_i \beta_i > 0$	$\alpha_i \beta_i < 0$	$\alpha_i = 0$	$\beta_i = 0$
0	0	False	0	0	0	0	0
0	0	True	0	0	0	0	0
0	1	False	0.005π	+1	-1	0	± 1
0	1	True	0.025π	-1	+1	± 1	0
1	0	False	0.05π	-1	+1	± 1	0
1	0	True	0.025π	+1	-1	0	± 1
1	1	False	0.005	+1	-1	0	± 1
1	1	True	0.025π	+1	-1	0	± 1

Table 3. Example of Chromosome measurement of 3 genes

	Gene 1		Gene 2		Gene 3	
$(\alpha_{ij})^2$	0.71	0.42	0.29	0.58	0.71	0.71
$(\beta_{ij})^2$	0.29	0.58	0.71	0.42	0.29	0.29
$r(r \text{ is random } 0 < r < 1)$	0.6	0.45	0.5	0.7	0.3	0.6
Test of condition	$r \geq \alpha ^2$	$r < \alpha ^2$	$r < \alpha ^2$	$r < \alpha ^2$	$r \geq \alpha ^2$	$r \geq \alpha ^2$
Binary value of α_{ij}^t	0	1	1	1	0	0
Binary value of β_{ij}^t	1	0	0	0	1	1

Table 4. CQM Care Qualification Matrix of care workers

	T1	T2	T3	T4	T5
CW1	1		1		1
CW2		1	1	1	
CW3	1	1		1	1
CW4	1	1	1	0	0

Table 5. MPR Matrix of patient requirements

	P1	P2	P3	P4	P5	P6	P7	P8	P9
T1	1		1						
T2	1	1		1					
T3		1	1			1			
T4					1		1		
T5								1	1

Table 6. Symmetric Matrix of distances between patients

	C	P1	P2	P3	P4	P5	P6	P7	P8	P9
C	0	7.28	7.81	4.47	5	5	7.07	6.71	4	1.41
P1	7.28	0	2.2	3	4	3	3.6	5.7	7.1	6.1
P2	7.81	2.24	0	4	3	4	1.4	3.6	6	5.8
P3	4.47	3	4	0	2	3	4.1	6	4.5	2.2
P4	5	4.47	3	2.2	0	4	2.2	2	4	3.6
P5	5	2.83	4.5	3.2	4	0	5	6.3	6.4	4.1
P6	7.07	3.61	1.4	4.1	2	5	0	2.2	5.1	5.4
P7	6.71	5.66	3.6	6	2	6	2.2	0	3.6	5.4
P8	4	7.14	6	4.5	4	6	5.1	3.6	0	3.2
P9	1.41	6.08	5.8	2.2	4	4	5.4	5.4	3.2	0

Fig. 1. Quantum chromosome structure (5,16).

Fig. 2. Example of Quantum chromosome structure with size $k=2$.

Fig. 3. Quantum Rotation Gate Qubit Update Operation (24)

Fig. 4. Sample of possible routing of 4 care teams

Fig. 5. Conventional GA chromosome coding

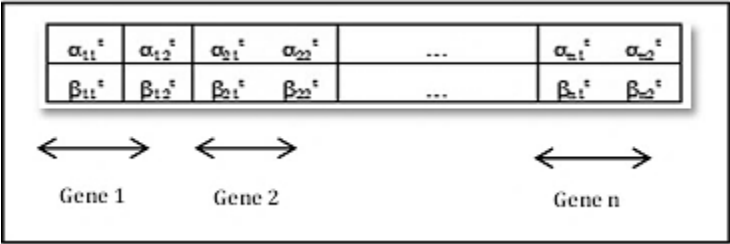
Fig. 6. Quantum GA chromosome coding

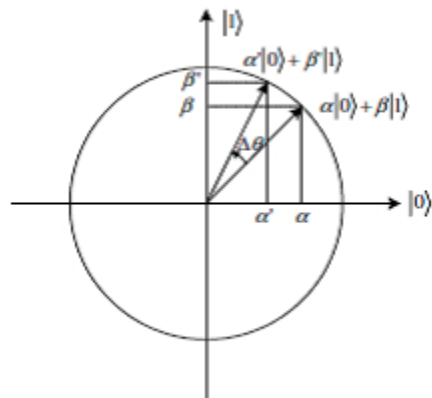
Fig. 7. Chart of Quantum Inspired Genetic Algorithm

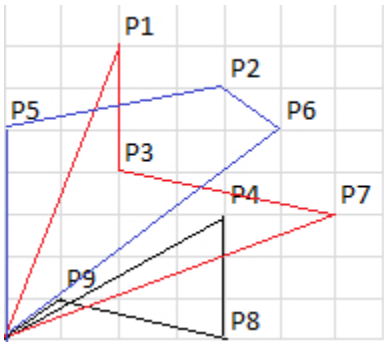
Fig. 8: Comparison between GA and QIGA

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$$q_j^t = \left[\begin{array}{c|c|c|c|c|c|c|c|c|c} \alpha_{11}^t & \alpha_{12}^t & \cdots & \alpha_{1k}^t & \alpha_{21}^t & \alpha_{22}^t & \cdots & \alpha_{2k}^t & \cdots & \alpha_{n1}^t & \alpha_{n2}^t & \cdots & \alpha_{nk}^t \\ \hline \beta_{11}^t & \beta_{12}^t & \cdots & \beta_{1k}^t & \beta_{21}^t & \beta_{22}^t & \cdots & \beta_{2k}^t & \cdots & \beta_{n1}^t & \beta_{n2}^t & \cdots & \beta_{nk}^t \end{array} \right]$$







1	2	1	3	2	2	1	3	3	3	3	3										
0	1	1	0	0	1	1	1	1	0	1	0	0	1	1	1	1	1	1	1	1	1

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